

GARUDA

(Web application)

**FIELD WISE EXPLANATION
FOR GOOGLE SHEET COLUMNS**

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I. BASIC IDENTIFICATION MODULE

A. Sl. No. (Serial Number)

1.Importance

- Unique internal reference number for each suspect/consumer.
- Prevents duplication of records.
- Enables linking of records across PSs and cases.

2.Expected Analytics

- Total suspects registered.
- Year-wise / PS-wise growth trends.
- Repeat offender identification.
- Record duplication detection.

3.Intelligence Use

- Creation of centralized district-level NDPS registry.
- Quick retrieval during investigations.

B. PS (Police Station)

1.Importance

- Identifies jurisdiction of the individual.
- Enables territorial crime mapping.

2.Expected Analytics

- PS-wise NDPS hotspots.
- PS-wise arrested accused statistics.
- High-risk police station identification.
- Emerging drug-prone police station analysis.
- One accused resident of one place / jurisdiction committing of crime in another jurisdiction.

3.Preventive Intelligence

- Resource allocation based on drug activity.
- Targeted surveillance in high-incidence PS limits.
- Deployment of special teams.

4.Predictive Intelligence

- Forecasting future NDPS activity areas.
- Identifying migration of peddlers between PS limits.

C. Name of Consumer / Suspect

1.Importance

- Core identity field.
- Enables linking with FIRs, CDRs, bank accounts, social media, and intelligence databases.
- Allocation of Unique ID for each accused.

2.Expected Analytics

- Repeat involvement patterns.
- Alias-based duplicate detection.
- Network linkage between accused.
- One accused resident of one place / jurisdiction committing of crime in another jurisdiction.

3.Intelligence Use

- Association mapping.
- Criminal network visualization.

4.Predictive Use

- Identification of recurring offenders likely to re-engage in NDPS activity.

D. Test Results (Positive / Negative)

1.Importance

- Differentiates actual consumers from suspects.
- Scientifically validates consumption.

2.Expected Analytics

- Percentage of confirmed consumers.
- Area-wise drug consumption trends.
- Drug prevalence ratio.

3.Preventive Intelligence

- Identification of emerging consumption zones.
- Monitoring of schools/hostels/colleges if clusters emerge.
- For proceeding further to extract information about hierarchy (local seller) of drug offenders in the area.

4.Predictive Intelligence

- Forecasting increase in drug abuse in specific localities.

E. Category / Hierarchy of Suspect

1.Categories

- Consumer
- Local Peddler
- Supplier
- King Pin
- Transporter
- Interstate King Pin

2.Importance

This is one of the most important intelligence fields.

3.Expected Analytics

- Drug supply chain hierarchy mapping.
- From consumer proceeding further to extract information about hierarchy (local seller) of drug offenders in the area.
- Identification of distribution networks.
- Detection of operational structures.
- Mapping of cell phone connections, conversations and connections.
- Mapping of cell phone IMEI device connections, conversations and connections.
- Mapping of bank accounts / transactions connections.
- Mapping of UPI accounts / transactions connections
- Mapping of Jio locations
- Mapping of vehicles used

4.Preventive Intelligence

- Focused action against all ranks of hierarchy like suppliers etc. rather than only consumers.
- Identification of source-to-street distribution channels.
- Detection of interstate trafficking routes.

5.Predictive Intelligence

- Prediction of:
 - Upcoming supply movements
 - Expansion of peddler networks
 - New distribution zones

6.Advanced Intelligence Possibilities

- Network graph analytics.
- Kingpin influence mapping.
- Automated hierarchy detection.

F. Father / Husband Name

1.Importance

- Assists in identity verification.
- Useful in tracing family-linked offenders.
- For establishing connections through cell phones
- For establishing connections through IMEI
- For establishing connections through bank account and transactions.
- For establishing connections through UPI accounts and transactions
- For establishing connections through vehicles involved /seized

2.Expected Analytics

- Family-based criminal involvement patterns.
- Repeat involvement within households.

3.Preventive Intelligence

- Detection of family-supported trafficking operations.

G. Age

1.Importance

- Identifies vulnerable age groups.

2.Expected Analytics

- Age-wise consumption trends.
- Youth involvement analysis.
- Student-age drug abuse mapping.

3.Preventive Intelligence

- School/college awareness targeting.
- Youth rehabilitation planning.

4.Predictive Intelligence

- Forecasting drug abuse spread among specific age categories.

H. Gender

1.Importance

- Helps understand demographic spread.

2.Expected Analytics

- Male/Female/Others involvement ratio.
- Gender-specific trends.

3.Preventive Intelligence

- Women-focused counselling/intervention programs.
- Identification of women courier networks.

II. ADDRESS DETAILS MODULE

A. Full Address

1.Importance

- Critical for geospatial intelligence.
- Enables mapping and surveillance.

2.Expected Analytics

- Street-wise hotspot mapping.
- Colony-wise drug prevalence.
- Repeat-offender concentration areas.

3.Preventive Intelligence

- Area domination operations.
- Focused patrolling.
- Surveillance installations.
- CCTV surveillance
- Drone surveillance
- Cordon and search operations

4.Predictive Intelligence

- Identification of emerging drug clusters.
- Crime forecasting based on locality history.

5.Advanced Capabilities

- GIS-based heat maps.
- Geo-fencing alerts.
- Distance analysis from schools/hostels.

B. Landmark / Area

1.Importance

- Helps operational teams identify exact zones quickly.
- Geo locations coordinates – tracking

2.Expected Analytics

- High-risk landmarks.
- Public places vulnerable to drug activity.

3.Preventive Intelligence

- Monitoring:
 - Bus stands
 - Railway stations
 - Hostels
 - Colleges
 - Markets
 - Cinema halls
 - Play grounds
 - Shandy places
 - Fairs & Festivals
 - Isolated areas

4.Predictive Intelligence

- Predicting congregation points for consumers and peddlers.

C. District

1.Importance

- Enables district-wise intelligence integration.

2.Expected Analytics

- Inter-district movement patterns.
- Source districts for supply.

3.Preventive Intelligence

- Coordination with neighboring districts.

4.Predictive Intelligence

- Forecasting incoming trafficking routes.

D. State

1.Importance

- Crucial for interstate trafficking analysis.

2.Expected Analytics

- State-wise supply chain origin mapping.
- Interstate trafficking corridor identification.

3.Preventive Intelligence

- Coordination with other state police agencies.
- Interstate surveillance alerts.

4.Predictive Intelligence

- Prediction of future trafficking routes.

III. HIGH-END ANALYTICS POSSIBLE THROUGH THE WEB APPLICATION

A. Crime Network Analytics

The system can automatically identify:

- Suppliers linked to multiple consumers.
- Common transporters.
- Shared mobile numbers.
- Shared addresses.
- Financial transaction links.

Output

- Criminal network diagrams.
- Syndicate structures.
- Hidden associations.

B. Geographic Intelligence (GIS Analytics)

Possible Outputs

- Heat maps of NDPS activity.
- Route mapping.
- Hotspot analysis.
- Colony-wise prevalence.

Preventive Use

- Dynamic patrolling plans.
- Check-post deployment.

C. Predictive Policing

Using historical data, the system may predict:

- Future hotspot areas.
- Repeat offenders.
- Likely trafficking routes.
- Emerging drug markets.

Alerts

- New accused entering hotspot areas.
- Multiple consumers linked to one address.
- Frequent interstate movement.

D. Behavioral Intelligence

By integrating:

- Cell phone data
- Geo location
- CDR analysis
- Social media

The system can identify:

- Meeting patterns.
- Night movement behavior.
- Frequent contact analysis.

E. Financial Intelligence

After integrating:

- Bank accounts
- UPI IDs
- Transaction history

The system can detect:

- Suspicious repetitive payments.
- Small-value layering transactions.
- Financial links between peddlers and consumers.

Advanced Outputs

- Money flow mapping.
- Illegal transaction clusters.

IV. AUTOMATED ALERTS POSSIBLE

The system can generate automatic alerts for:

- a) Repeat offenders.
- b) Movement into sensitive zones.
- c) Interstate accused entry.
- d) Multiple accused using same address/mobile/IMEI/UPI/Bank A/C.
- e) Sudden increase in consumers in one locality.
- f) Newly emerging peddlers.
- g) Hotspot escalation alerts.

V. STRATEGIC BENEFITS TO POLICE

1.Operational Benefits

- Faster investigation.
- Centralized NDPS intelligence.
- Better coordination between PSs.

2.Supervisory Benefits

- District-level monitoring dashboard.
- PS performance analytics.
- Trend-based deployment planning.

3.Preventive Benefits

- Early intervention.
- Youth protection.
- Targeted surveillance.

4.Intelligence Benefits

- Kingpin identification.
- Supply chain disruption.
- Interstate coordination.

VI. FUTURE EXPANSION POSSIBILITIES

The application can later integrate:

- a) Facial recognition
- b) FP Bio-metric recognition
- c) Aadhaar verification
- d) Vehicle movement analytics
- e) CCTV integration
- f) Drone surveillance inputs
- g) AI-based risk scoring
- h) Predictive offender ranking
- i) Prison intelligence
- j) Court case tracking
- k) Rehabilitation monitoring

VII. SUGGESTED DASHBOARDS

The web application should ideally contain:

- a) District NDPS Dashboard
- b) PS-wise Dashboard
- c) Hotspot Map Dashboard
- d) Repeat Offender Dashboard
- e) Most wanted criminals
- f) Supply Chain Dashboard
- g) Financial Intelligence Dashboard
- h) Mobile/CDR Intelligence Dashboard
- i) Interstate Network Dashboard
- j) Predictive Risk Dashboard
- k) Surveillance Alert Dashboard
- l) Key resources (PDFs of Law books, training material, SOP, GOs, circulars and memos)